

ROHIT RATHOD

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PROFESSIONAL SUMMARY

Motivated Data Science graduate (CGPA: 8.29) with hands-on experience in Python, SQL, machine learning, and data analytics through internships and 6+ projects. Proficient in building and deploying end-to-end ML pipelines using Scikit-learn, TensorFlow, and Docker. Experienced in NLP (sentiment analysis, TF-IDF), Computer Vision (YOLOv5, CNN, OpenCV), and Big Data tools (PySpark, Spark). Skilled in data visualization with Power BI and Tableau, and cloud deployment on AWS and Azure. Eager to contribute data-driven insights and scalable AI solutions to a dynamic organization as a Junior Data Scientist or Data Analyst.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Bash

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, MS Excel (Pivot Tables, VLOOKUP)

Machine Learning & AI: Scikit-learn, TensorFlow, Keras, CNN, NLP, Regression, Classification, Clustering, EDA, Feature Engineering

Computer Vision: YOLOv5, OpenCV, Grad-CAM, Image Processing

Big Data & Streaming: Apache Spark, PySpark, Kafka (Basics)

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, Terraform, CI/CD

Databases: MySQL, PostgreSQL

APIs & Deployment: Flask REST API, Model Serving

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Linux CLI

INTERNSHIP EXPERIENCE

Python Developer Intern | Pune, Maharashtra

- Automated analytics and reporting workflows using Python, Pandas, NumPy, and SQL, reducing manual data processing effort by 30%.
- Designed reusable data preprocessing pipelines for structured datasets comprising 500+ records, improving data quality and consistency.
- Built a Real-Time Fake Logo Detection System using YOLOv5, OpenCV, TensorFlow, and CNN with live webcam inference achieving 90% detection accuracy.
- Integrated Grad-CAM explainability heatmaps for model transparency and debugging, enhancing stakeholder trust in AI outputs.
- Containerized the detection system using Docker and prepared deployment-ready workflows for production use.

Data Science Intern | Pune, Maharashtra

- Implemented an NLP-based sentiment analysis pipeline on a social media dataset of 1,000+ tweets using TF-IDF, tokenization, Logistic Regression, and SVM.
- Performed Exploratory Data Analysis (EDA), text preprocessing (lemmatization, stop-word removal), and model evaluation using accuracy, precision, recall, and F1-score.
- Delivered actionable analytical insights and visualizations to support data-driven business decision-making.

PROJECTS

Real-Time Fake Logo Detection System

- Developed an end-to-end AI system using YOLOv5 + CNN to detect and classify logos from live webcam streams with 90% detection accuracy on 600+ training images.
- Implemented OpenCV frame processing pipeline and TensorFlow/Keras model inference for real-time classification.
- Added Grad-CAM heatmaps for explainability; containerized using Docker for deployment-ready delivery.

Twitter Sentiment Analysis | NLP Pipeline

- Built a full NLP pipeline to classify tweets into positive, negative, and neutral categories using TF-IDF vectorization and ML models (Logistic Regression, SVM).
- Applied text cleaning, lemmatization, and evaluated model performance using confusion matrix and F1-score metrics.

Customer Segmentation System | K-Means Clustering

- Segmented customers based on purchasing behavior using K-Means clustering; applied PCA for dimensionality reduction and 2D visualization.
- Generated actionable business insights for targeted marketing and customer retention strategies.

BigMart Sales Prediction | Regression Modeling

- Conducted EDA and feature engineering on retail sales data; built Linear Regression and Decision Tree models to forecast product-level sales.
- Identified key business drivers impacting sales performance, improving prediction reliability for business planning.

AI-Driven SQL Query Optimizer | ML + SQL

- Designed a system to analyze SQL queries and recommend index optimizations and query rewrites using extracted features (joins, filters, table scans).
- Built an ML model to predict query execution time, improving database query performance for large datasets.

ML Model Deployment with Docker & Flask

- Built Flask REST APIs for serving trained ML models and containerized applications using Docker for a CI/CD-style deployment workflow.

EDUCATION

B.Tech in Information Technology (Data Science) | CGPA: 8.29 | 2025

Ajeenkya DY Patil University, Pune

Higher Secondary Certificate (Science) | 86.67% | 2021

Royal Junior College of Science, Thane

CERTIFICATIONS

- SQL for Data Science – University of California, Davis (Coursera)
- Databases and SQL for Data Science with Python – IBM (Coursera)
- AWS Training & Cloud Fundamentals – Infosys
- Excel Skills for Business – Macquarie University (Coursera)

KEY STRENGTHS

Data Analysis • Machine Learning • Statistical Analysis • Data Visualization • Problem Solving • EDA • Model Deployment • Attention to Detail • Cross-functional Collaboration • Continuous Learning

LANGUAGES

English • Hindi • Marathi